# FD E-Commerce Analytics Dashboard

Interactive intelligence for sales, customer preferences, marketing ROI and strategic decisions

## FROM SCATTERED TRANSACTIONS TO MEASURABLE BUSINESS DECISIONS

Python · MongoDB · Streamlit · Pandas · Plotly

Prepared by Jagjot Singh Passi | BCA | Course Instructor: Ms. Sarbjeet Kaur

# What business problem are we solving?

Three official academic requirements drive the entire design.

01 **Sales trends**

Find seasonal movements, high-revenue products and regions.

02 **Customer preferences**

Measure repeat buying, customer segments and product mix.

03 **Marketing return**

Link campaign spend with attributed orders and report ROAS / ROI proxy.

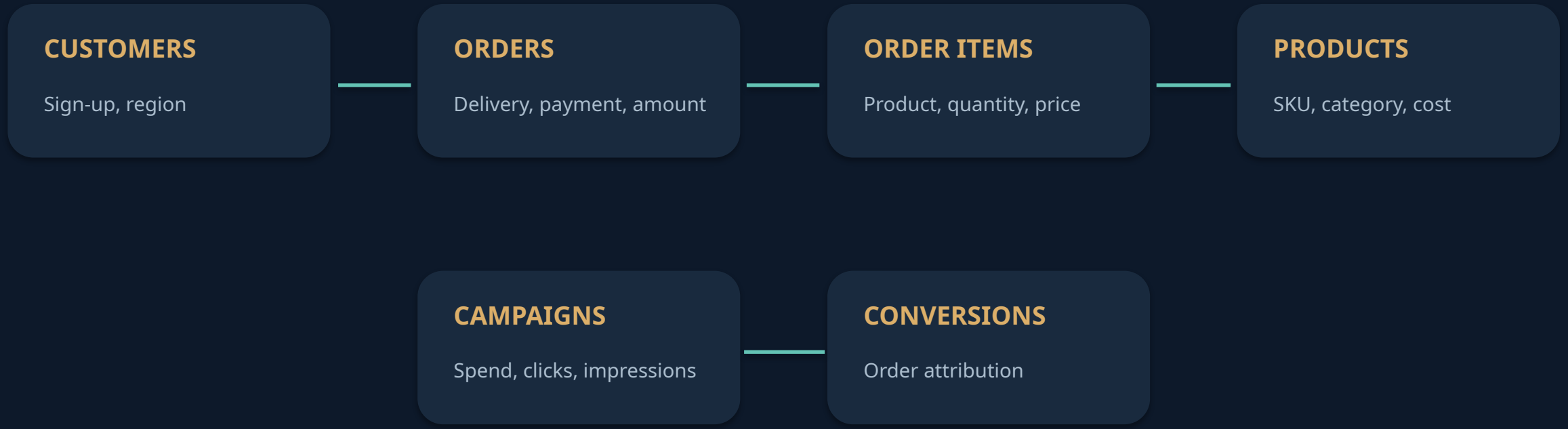
# Implementation stack & architecture

Matches the Phase 1 evaluated stack; offline demo fallback is always labeled.

| | |
|---|---|
| DATA SOURCES | Authorized CSV/Excel exports or reproducible simulation |
| VALIDATION | Six linked tables, arithmetic and referential checks |
| PERSISTENCE | MongoDB collections · SQLite local demo mode |
| ANALYTICS | Pandas KPI formulas, RFM, cohorts and attribution |
| FRONTEND | Streamlit controls + Plotly interactive charts |

# Six related collections

Customer → order → order item → product; campaign → attributed delivered order




Validated unique IDs, foreign keys, reconciliation and single-touch delivered-order attribution.

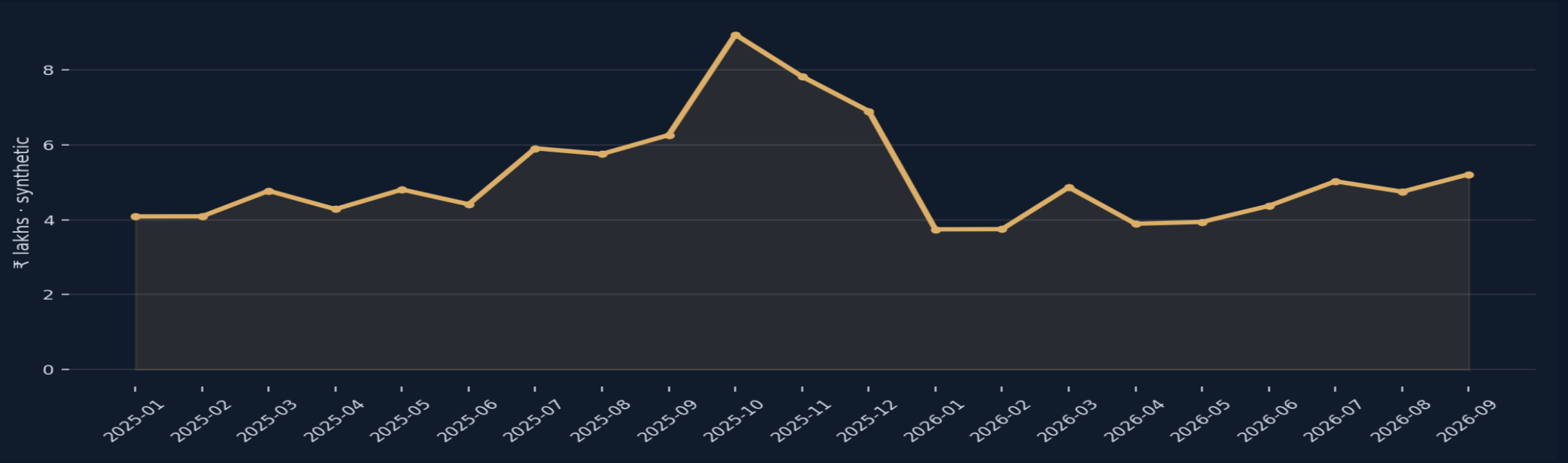
# Delivered revenue changes with filters

Date, region and category selection recalculate qualifying line-level revenue.

**₹1.08 Cr**
Synthetic realized sales

**4,399**
Delivered orders

**₹2,444**
Average selected order



# From transactions to customer segments

RFM + cohort retention + exploratory clustering are different lenses on buying behavior.

01 **RECENCY**
Days since last delivered purchase

02 **FREQUENCY**
Number of delivered purchases

03 **MONETARY**
Sum of delivered spending

**891**
Customers with delivered history

**77.4%**
Repeat-buyer share

# Attribution & marketing measurement

Reported national marketing spend is prorated for partial campaign date ranges.

**₹6.39 L**

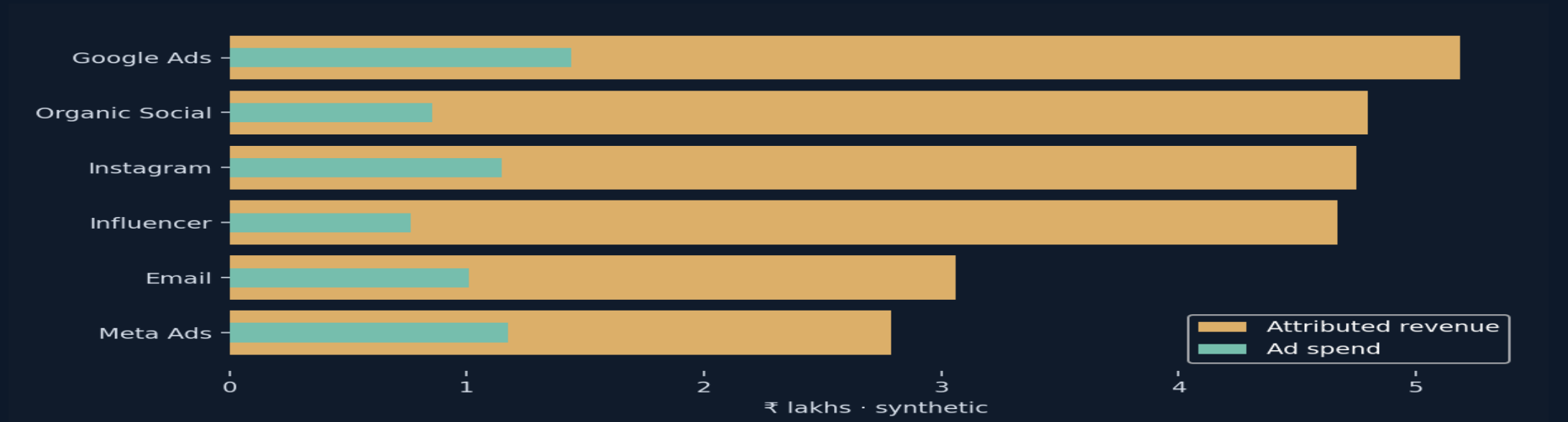
Advertising spend · synthetic

**₹25.25 L**

Attributed sales · synthetic

**3.96×**

Single-touch ROAS



# Know what the formulas mean

Never confuse attributed sales with verified incremental profit.

| | |
|---|---|
| **Revenue** | Σ (delivered item quantity × sale price) |
| **ROAS** | Attributed delivered-order revenue ÷ campaign spend |
| **CPA** | Campaign spend ÷ attributed delivered orders |
| **ROI proxy** | (Attributed revenue − estimated product COGS − spend) ÷ spend |
| **RFM** | Recency · Frequency · Monetary based on delivered-order history |

# Dynamic imports, validation and audit trail

The application rejects broken datasets before activating their records.

| | |
|---|---|
| **INPUT** | One workbook (six sheets) or six CSV files |
| **QUALITY** | Unique IDs, balanced totals, valid dates and status rules |
| **DATABASE** | MongoDB batch switch or SQLite transactional replacement |
| **OUTPUT** | Dynamic dashboards, CSV/Excel exports and backup ZIP |
| **AUDIT** | Import origin + timestamp + administrator action log |

# 10 automated checks completed

Tests were executed for the analytics core and SQLite database persistence.

01 **Sales logic**
Reconciliation, cancellation exclusion, category sums

02 **Marketing logic**
Single-touch aggregation, spend prorating

03 **Customer logic**
RFM labeling and cohort month 0

04 **Data quality**
Reject incorrect orders and attribution

05 **Persistence**
SQLite roundtrip, metadata and failed-validation safety

# Real-world data: what is available and what is not

A simulation can demonstrate software, but it must never be presented as measured business activity.

## PROVIDED NOW

Reproducible, internally consistent synthetic records, realistic prices and seasonality, documented provenance.

## NEXT DATA SOURCE

Permissioned store exports, anonymized customers and authentic ad campaign logs with order-level attribution.

**No actual FD-store transactions or live Shopify/webhook integration are claimed.**

# A tested business intelligence prototype

Demonstrate: database persistence → live filter changes → sales → RFM → marketing → import/export.

01 **Meets the analytical scope of the faculty brief**

02 Upgrades the existing Python prototype with persistent database support

03 Provides explicit KPI definitions and source-lineage disclosure

04 Ready for authorized real exports and future live-platform integration

**Thank you • Questions & demonstration**